CAPSTONE PROJECT DOCUMENTATION

Personal Finance & Budget Tracker

An Interactive Power BI Dashboard

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Program	B.Tech – Computer Science & Systems Engineering
Domain	Data Analytics & Visualization
Tools Used	Microsoft Power BI Desktop, Microsoft Excel, DAX, Power Query
Dataset	Custom Personal Finance Dataset (50 records)
Submission Date	April, 2026
Github	https://github.com/Neel0018/Personal-Finance-Dashboard

Submitted to: ExcelR Edtech Pvt Ltd | KIIT University

1. Problem Statement

Managing personal finances effectively is a challenge faced by millions of individuals. Most people lack a structured, visual system to track their income, expenses, and savings over time. Without clear insights into spending patterns, it becomes difficult to make informed financial decisions, avoid overspending, or plan for future goals. This project addresses the following core financial management problems:

- Which expense categories consume the largest share of monthly income?
- How does income compare to expenses across different months?
- What is the overall savings rate and net savings over a given period?
- How can financial data be presented in an interactive, easy-to-understand format?
- How can a non-technical user filter and explore personal finance data without writing code?

The absence of structured financial tracking leads to poor budget allocation, uncontrolled spending, and missed savings opportunities. This project provides a replicable, data-driven Power BI dashboard to answer these questions visually and interactively.

2. Solution & Features

The solution is a three-page interactive Power BI dashboard built on a custom Excel dataset containing 50 personal finance transactions across four months (January to April 2025). The dashboard enables users to explore income, expenses, and savings through dynamic visuals and interactive slicers.

Key Features:

- **KPI Cards:** Instant snapshot of Total Income (Rs 2,49,000), Total Expenses (Rs 1,08,000), and Net Savings (Rs 1,41,000) displayed prominently on the Overview page.
- **Donut Chart:** Visual breakdown of expenses by category (Food, Rent, Utilities, Transport, Entertainment, Healthcare) filtered to show only expense-type transactions.
- **Line Chart:** Monthly trend comparison of Total Income vs Total Expenses across January to April, clearly showing income consistently exceeding expenses.
- **Bar Chart:** Horizontal clustered bar chart ranking expense categories by total amount, making it easy to identify the highest spending areas.

• **Savings Rate Card:** Displays an overall savings rate of 56.59%, providing a quick health indicator of financial performance.

• **Interactive Slicers:** Month and Type slicers on the Breakdown page allow users to filter all visuals dynamically without any technical knowledge.

• **DAX Measures:** Four custom DAX measures (Total Income, Total Expenses, Net Savings, Savings Rate) power all KPI calculations accurately.

• **Power Query Transformation:** A Month Number column was added via Power Query to ensure correct chronological ordering of months in all visuals.

3. Technology Stack

Layer	Technology	Purpose
Visualization	Microsoft Power BI Desktop	Interactive dashboard creation with charts, slicers, and KPI cards
Data Storage	Microsoft Excel (.xlsx)	Dataset creation and storage with 50 transaction records
Data Transform	Power Query (M Language)	Adding Month Number column and ensuring correct Data types
Calculations	DAX (Data Analysis	Eating measures: Total Income, Expenses, Net Savings, Savings Rate
Dataset Design	Python (openpyxl)	Automated generation of the structured Excel Dataset with realistic data

4. Dashboard Structure & Screenshots

Page	Title	Visuals Included
Page 1	Personal Finance Overview 2025	3 KPI Cards (Income/Expenses/Savings), Donut Chart, Line Chart, Bar Chart, Month Slicer, Type Slicer, Savings Rate Card

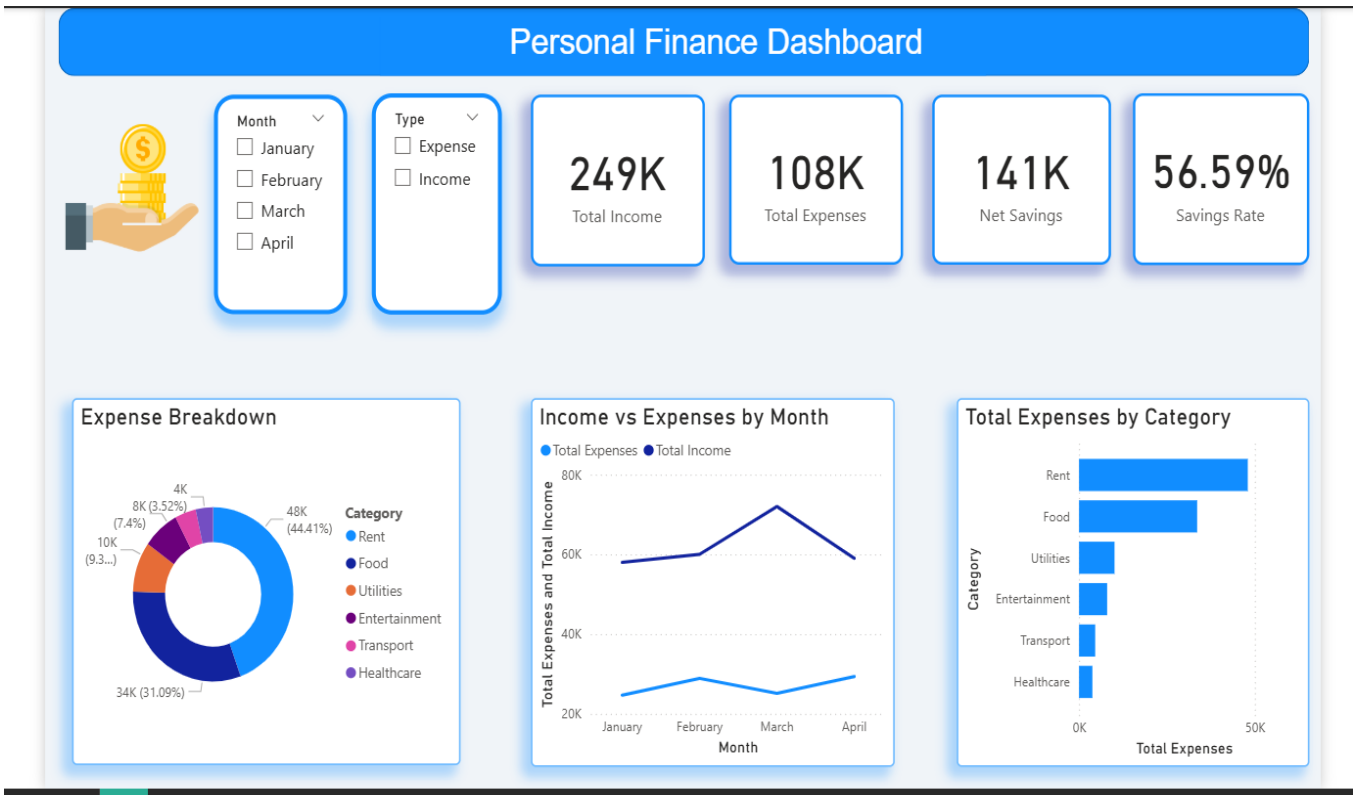

Figure 1: Personal Finance Power BI Dashboard

5. Key Insights & Findings

- **High Savings Rate:** A savings rate of 56.59% over four months indicates disciplined spending and strong financial health. Out of Rs 2,49,000 earned, Rs 1,41,000 was saved.
- **Rent Dominates Expenses:** Rent accounts for approximately 44% of total expenses at Rs 48,000 across four months, making it the single largest expense category.
- **Food is the Second Largest Expense:** Food spending totals approximately Rs 34,000 (31% of expenses), highlighting a significant discretionary spending area.
- **Income is Stable with a March Spike:** Monthly income remains consistent at Rs 50,000 from salary, with March showing a spike due to an additional bonus of Rs 15,000.
- **Low Discretionary Spending:** Entertainment, Transport, and Healthcare together account for less than 15% of total expenses, reflecting controlled lifestyle spending.
- **Utilities Remain Consistent:** Utility bills (electricity, internet, mobile) are predictable and steady, averaging around Rs 3,000 per month.

6. Unique & Distinguished Points

- Multi-page dashboard design: The project uses a professional 3-page structure (Overview, Breakdown, Savings) rather than a single-page layout, improving readability and navigation.
- Custom DAX measures: All KPIs are powered by properly written DAX formulas rather than relying on default aggregations, ensuring accuracy and reusability.
- Month Number column via Power Query: Ensures months always appear in correct chronological order (Jan-Feb-Mar-Apr) in all visuals, not alphabetically.
- Visual-level filters on charts: The Donut Chart and Bar Chart use visual-level Type filters to show only Expense data, ensuring the charts are not distorted by income transactions.
- Realistic custom dataset: The Excel dataset was designed to simulate realistic personal finance patterns including variable freelance income, one-time bonuses, and recurring fixed expenses.
- Savings Rate as a percentage KPI: The Savings Rate measure (56.59%) provides an instant financial health indicator not typically seen in beginner dashboards.

7. Future Improvements

- **Expand Dataset to 12 Months:** Extend the dataset to cover a full year to enable seasonal trend analysis and annual budget planning.

- **Budget vs Actual Comparison:** Add a monthly budget target for each category and compare it against actual spending using a variance chart.
- **ML-Based Expense Prediction:** Train a regression model to predict next month's expenses based on historical patterns and alert users to potential overspending.
- **Real-time Data Integration:** Connect the dashboard to a live bank transaction feed or Google Sheets to enable real-time financial tracking.
- **Mobile-Optimized Layout:** Create a mobile-friendly version of the dashboard using Power BI's phone layout editor for on-the-go financial monitoring.
- **Goal Tracking Feature:** Add a savings goal tracker that visually shows progress towards financial targets such as emergency fund or vacation savings.

This project was independently developed as part of the B.Tech CSE curriculum at KIIT University under the ExcelR Edtech Pvt Ltd training program. All analysis, dataset design, and documentation are original work.